

Special Characteristics and L-Marking

Identification, designation and handling of Special
Characteristics and L-marking

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Continued page 2 to 33

Contents

1	Scope and purpose	3
2	Normative references	3
3	Abbreviations, terms, definitions and symbols	3
3.1	Abbreviations.....	3
3.2	Terms.....	4
3.3	Symbols.....	5
4	Special Characteristics	6
4.1	Categories of Special Characteristics	6
4.2	Consequences of designation characteristics as Special Characteristics	7
5	L-marking and BM-L.....	8
5.1	Specification of L-marking and type approval-relevant BM-L.....	8
5.2	Application of L-marking and type approval-relevant BM-L in change management.....	8
6	Identification of Special Characteristics	8
6.1	Identification of Special Characteristics by means of FMEA	8
6.2	Identification of Special Characteristics without an FMEA.....	11
6.3	Explanations on the identification of Special Characteristics.....	13
7	Application of Special Characteristics in production.....	13
7.1	Defining process measures.....	13
7.2	Handshake with downstream processes	13
8	Designating Special Characteristics and L-marking	14
8.1	Denoting Special Characteristics in summarizing tables	14
8.2	Designating L-marking and type-approval relevant BM-L in quantity structures and change management systems	16
8.3	Designation in drawings, graphics and 3D models.....	16
8.4	L-marking on the cover sheet of documents.....	18
9	Documentation and archiving.....	18
A	(Annex informative) Technical categories of Special Characteristics	19
B	(Annex normative) Filters	20
C	(Annex informative) Example decomposition	23
D	(Annex informative) Overall view concept/design/production filters	24
E	(Annex normative) Severity S and rating index BI	25
F	(Annex normative)	31

Foreword

The German version is binding.

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Amendments

The following amendments have been made to GS 91011:2019-08:

- Section 3.2, 5 (is now section 4), 6, 7, 8, 9 updated;
- Annex C, E.3, E.8 and F updated;
- section 3.3 4 removed;
- normative references updated;
- Group Standard editorially revised.

Previous editions

BMWN(S) 11369.0: 1993-11, 1996-06, 1997-03

GS 91011: 2008-05, 2009-03, 2012-04, 2015-06, 2019-08

1 Scope and purpose

This Group Standard governs the identification, designation and handling of Special Characteristics and L-marking.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

GS 90017	Continuous validation of product reliability in the series phase; Process description
GS 90036	Identification of parts; Rules for definition and implementation of traceability of parts
GS 91001	Marking of parts; with trademark and part identification data
GS 91004	Marking of parts; Marking with date of manufacture
GS 91010-1	Component identification marking; Marking with barcode for IPS-Q, requirements
GS 91010-2	Marking of parts; Marking with Data Matrix Code or RFID label for IPS-Q, requirements
GS 98000	Statistical capability studies
GS 98003	FMEA; Failure Modes and Effects Analysis
UN R156	UN Regulation No. 156; Software update and software update management system
VDA Volume Special Characteristics	Joint Quality Management in the Supply Chain; A process description covering Special Characteristics (SC)
AIAG-VDA-FMEA Handbook	Failure Mode and Effects Analysis; Design FMEA, Process FMEA, Supplemental FMEA for Monitoring & System Response)

3 Abbreviations, terms, definitions and symbols

3.1 Abbreviations

AIDA	Architecture, integration and design for automotives
S	Severity
BFF	Special failure impact
BM	Special characteristic
DFMEA	Design FMEA
DoE	Design of Experiments (statistical test planning)
E/E	Electrical system/electronics
EoL	End of Line
ET	Single part
ETK	Electronic parts catalog
EVE	Determination of consumption and emissions
FL	Function list
FMEA	Failure Mode and Effects Analysis
IP	Implementation planning
JIS	Just in sequence
PEVFA	Advanced technical development specialist departments
LSV	Service Interface Agreement
M-gAMS	Measures and approving change management system
MSA	Measuring system analysis
MSR	Monitoring and system response
OtD	Order to Delivery

PBM	Potential Special Characteristics
PEP-PDM	Product engineering process - product data management
PFMEA	Process FMEA
PLP	Production control plan
PRISMA	Product information system with archive
PV-D	process owner for derivative
QFD	Quality Function Deployment
S-AFO	Elemental operation statement
SPC	Statistical process control
SWL-C	Software logistics cockpit
TecSa	Technical cleanliness
UQL	Unacceptable quality level
VTF	Process owner type approval specialist fields (role in the type approval process workflow)
VTP	Process owner type approval projects (role in the type approval process workflow)
ASSY	Assembly
Drwg	Drawing

3.2 Terms

A/B/C classification	The A/B/C classification is a part risk assessment tool. A/B/C stands for a high, moderate or low risk class.
AIDA FL/IP(BMW internal)	Quantity structure for electrical system/electronics (E/E) functions. It contains a function list (FL) and the implementation planning (IP) associated with it
Severity S	The severity S represents the impact of the failure for the customer in 10 stages. The highest level is S = 10 which means there is a risk to life and limb. The severity S plays an important role in classifying the Special Characteristic. $S \geq 7$ typically applies for Special Characteristics. An overview is available in Annex E.
Special failure impact	Consequence of an failure mode with influence on product and application safety, legal requirements, image or high warranty costs.
Evaluation index BI (BMW internal)	Like the severity S, BI represents the impact of the failure for the customer in 10 levels, but in reverse order. The highest level is BI = 1 which means there is a risk of fatality. BI may be used as an alternative if no FMEA exists. Annex E provides an overview.
Build-to-print	Part from a supplier without development responsibility.
Filter	Concept, design and production filters exist. This means a predefined set of questions that may be used to reduce the Special Characteristics in the concept and design filter. In the production filter, on the other hand, the issue is whether a BM is relevant at the point of fulfillment (detailed description in Annex B and D).
L-marking	L-marking means marking components, functions and overall vehicle characteristics that have features for fulfilling type approval-relevant requirements. This marking uses the letter "L".

M-gAMS (BMW internal)	Measures and approving change management system (consistent system for processing, evaluating and approving measures, changes and their technical and economic effects).
Quantity structures	Listing of components, functions and overall vehicle properties in BMW internal systems such as PEP-PDM and AIDA FL/IP. In quantity structures, the L-marking of specific components, functions and overall vehicle properties is carried out as well as the designation of type approval-relevant Special Characteristics.
PEP-PDM (BMW internal)	System for the management of product data, including the generic part structure and other items.
Potential Special Characteristics	The prefix "potential" initially means a potential BM for which a decision will later be taken as to whether it will be adopted as a BM. This typically occurs in the DFMEA through the threat and risk analysis and/or after optimizations. The term PBM is used to differentiate this from previously defined Special Characteristic in the DFMEA, e.g. from predecessor projects.
PRISMA (BMW internal)	Central product data management system for geometry data in the product engineering process.
Production control plan PLP	Describes systems and processes required to control the production of a product. Among other things, it contains the documentation of all process steps, including tests and the defined procedure in case of deviations. This ensures that the Special Characteristic is actually checked in the on-going process.
Process parameters	In the production process, there are usually (machine) parameters that have a decisive influence on the desired product properties and are thus especially important for the Special Characteristic. One prerequisite for this is a process model (e.g. from a DoE) with which the relationships between process parameters and product properties may be predicted. This is particularly important when properties such as strength and weld quality may not be tested by non-destructive means.
RIMA (BMW internal)	RIMA is a risk management system that shows risks in the form of a matrix.
Robust characteristic	A characteristic is robust if a minor deviation from the specification does not significantly influence the function or a deviation from target value is improbable.
SWL-C (BMW internal)	In the context under analysis here: Approving system for E/E function changes.
Technical cleanliness	Freedom of a part or an assembly from dirt (contamination, chips, machining residues prior to putting into operation).
Reliability indicator (short rel. indicators acc. to GS 90017)	In contrast to the Special Characteristic, reliability indicators focus on the service life and not only on the manufacturing quality. They are broken down into rel. characteristics, rel. quality characteristics, rel. stress tests and rel. process parameters. The following applies: Rel. characteristics with a severity of $S \geq 7$ are always Special Characteristics. See GS 90017 for a detailed description.

3.3 Symbols

C_p	process or long-term capability
C_{pk}	process or long-term capability with mean value offset
n	random sample scope
USL	upper specification limit
p	probability
s	standard deviation
LSL	lower specification limit
$\bar{x}$	mean value

4 Special Characteristics

4.1 Categories of Special Characteristics

Table 1: Overview of Special Characteristics categories

Category	Criterion	Example
BM-L	Type approval (L)	– nitrogen oxide emissions quantity – retention values for passengers in case of a crash – sound-pressure level of the fanfare horn – brightness of the headlight
	other legal requirements (G)	– recycling quota – environmental requirements – laser light – radio frequencies
BM-S	Safety: impact on the use of the product, in particular critical driving situations caused by the product.	– unintended acceleration – faulty deceleration (malfunction of brake lines) – failure of power steering
	Health: impact on the health of the user or a third party.	– leak in fuel tank – service fluids escaping, e.g. because of the risk of flammability
BM-F	Functional requirements	– fit dimension at bearing seat
	technological requirement	– incorrect bolted connection of rear spoiler finisher (missing bolts)
	high economic loss	– water drain channel installed wrongly or not at all

BM-L Legal

BM-L are characteristics for components, functions and overall vehicle characteristics that result from requirements arising from legal requirements and whose non-compliance constitute violations of the law. They are differentiated according to the two use cases of type-approval-relevant BM and other legally relevant BM. In the summary table for BM (see section 8.1) the application case with a "type approval (L)" and "other legislation (G)" are listed in separate columns.

Application case "type approval (L)" (homologation):

BM-L (L) means, on the one hand, type approval-relevant characteristics of components, functions and complete vehicle characteristics, these are determined by the specialist unit "Type Approval, Development Abroad, Conformity of Production ". These characteristics are fundamentally immutable after the homologation maturity project milestone or may only be changed after repeating type approval.

NOTE Changing type-approval relevant characteristics of the L-marked components, functions and complete vehicle characteristics after the homologation maturity project milestone may lead to noncompliance with type approval legislation. Possible consequences include loss of homologation, Technical Campaigns, recalls mandated by authorities and penalties to companies.

Any changes to type-approval relevant features after the homologation maturity project milestone that are nevertheless necessary in the company's point of view may only be placed on the market after testing, documentation and, if necessary, official approval. This is ensured by applying the change management processes (in the M-gAMS, SWL-C systems).

Application case other legal requirements (G):

As a further aspect, the BM-L contain all characteristics that are necessary for fulfilling further legal requirements not relevant to type approval. These are defined and determined by the releasing specialist department itself.

NOTE Noncompliance with BM-L (G) leads to a loss of product conformity and to an associated defect of the vehicle that may lead to penalties and fines for the company.

BM-S Safety

BM-S are characteristics for the component, functional and complete vehicle characteristics, noncompliance with which may lead to a danger to life and limb and may endanger other road users and persons in a direct or indirect manner (operating safety and safety-in-use).

The releasing specialist department defines which characteristics and functions shall be marked as BM-S.

BM-F Function

BM-F are characteristics for the essential functions of components and complete vehicle characteristics, noncompliance with which may lead to high warranty costs (e.g. rework, cancelation of order, price reduction, damages) or rework. These include essential functional requirements, particularly for critical production processes such as tolerances, technological requirements and the 4 Fs (Form, Fit, Function, performance).

The releasing specialist department defines which characteristics and functions shall be marked as BM-F.

4.2 Consequences of designation characteristics as Special Characteristics

In development, characteristics shall be defined and identified as Special Characteristics. The consequences of this designation for production processes and for suppliers can be found in section 7. The exact place of implementation can be found in the summarizing table. When changing suppliers or production sites, the fulfilment of the BM shall be ensured, e.g. despite any other production concepts.

BM shall be documented in all pertinent documents and quantity structure (e.g. requirements specifications, drawings, 3D models, technical product descriptions or test and process planning documents, AIDA-FL, PEP-PDM, M-gAMS, SWL-C) (see section 9).

4.2.1 Ensuring the Special Characteristics in production

In production, the validation of the Special Characteristics shall be planned, carried out and passed according to the requirements of GS 98000.

4.2.2 Marking of parts with Special Characteristics

Parts with BM shall be marked according to GS 91001, GS 91004 and GS 91010-1 or GS 91010-2. Further methods are permissible in agreement with the responsible BMW planning departments (e.g. if the maximum daily volume defined for the serial number is insufficient).

4.2.3 Traceability of parts with Special Characteristics

Traceability according to GS 90036 shall be ensured for Special Characteristics.

4.2.4 Changes to Special Characteristics

Type approval-relevant characteristics shall not be changed after the homologation maturity project milestone; see section 5.

5 L-marking and BM-L

5.1 Specification of L-marking and type approval-relevant BM-L

L-marking and the content specifications for its type approval-relevant BM-L are defined by the specialist department "Type Approval, Development Non-Domestic, Conformity of Production". Based on global type approval legislation, the necessary characteristics are identified that may no longer be changed for the type-approved product.

The following also applies for L-marking of electrical system/electronics functions: it shall be used in combination with electrical system/electronics change management after the start of the type approval tests to ensure a robust process. The use of this process ensures that type approval requirements according to UN R156 and possibly other regulations, are met. The use of this process is therefore required for receiving type approval for software updates.

The L-marking and the BM-L (L) shall be checked and adapted when new type approval legislation is published or if the product structure (new or changed electrical system/electronics functions, new or changed generic components, new or changed complete vehicle properties) change. The process for adaptation/new assignment is shown as a schematic diagram in Annex F.1.

For BM-L that are not physically verifiable on a component or are influenced in the manufacturing process and are therefore not relevant to the production process, the place of implementation is given in the summarizing table "releasing specialist department BMW".

5.2 Application of L-marking and type approval-relevant BM-L in change management

L-marking makes it possible to process planned changes to marked components, functions and complete vehicle characteristics separately within the scope of change management in the BMW Group. Planned changes to L-marked scopes may be identified after the homologation maturity project milestone and specially tested for relevance to type approval. This makes it possible to ensure that only changes that are not relevant to type approval are released after the homologation maturity project milestone and that changes relevant to type approval are only implemented after type approval has been granted.

The processes in change management (in the M-gAMS and SWL-C systems) for testing a planned change after the homologation maturity project milestone are shown as schematic diagrams in Annex F.2 and F.3.

6 Identification of Special Characteristics

6.1 Identification of Special Characteristics by means of FMEA

The use of a FMEA is always the recommended method for identifying BM. If the system or component is classified as critical or safety-relevant by a threat and risk analysis, the FMEA shall be applied as a mandatory requirement (see A/B/C classification in Annex E.8).

To create an FMEA, the AIAG-VDA FMEA handbook and the GS 98003 shall be applied.

NOTE The advantage of a FMEA is structural consistency, especially by determining the significance of failure impacts of non-compliant functions and defective characteristics or properties. Basic knowledge of the FMEA is assumed in the further descriptions.

Within the scope of its due diligence, the supplier shall check the BMs specified by BMW and supplement them with further necessary BMs. This shall be documented in the appropriate FMEA and marked throughout. These changes and supplements shall be documented by the specifying specialist department and agreed upon with the releasing specialist department.

Figure 1 shows exemplarily the basic procedure for the identification of BM, its steps 1 to 9 are described below. The example shows a component drawing (Komp-Zchg.), usually a purchase part (e.g. airbag, gearbox, electric motor), as well as a single part drawing (ET-Zchg.) as an example of a production drawing or a subcomponent (e.g. anchor in the e-motor).

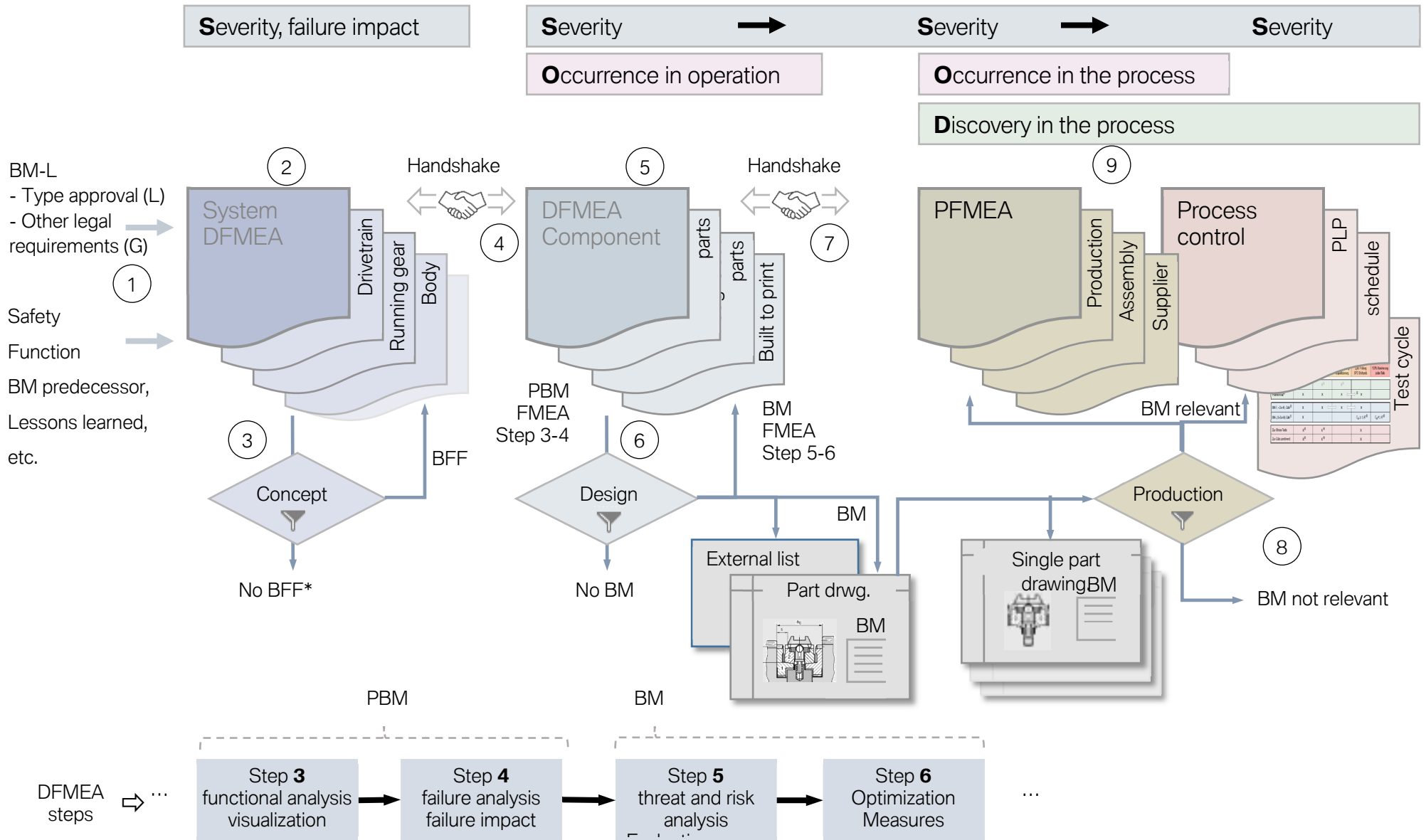


Figure 1: Complete process for identifying Special Characteristics with a FMEA

1 Requirements and specifications

Compile type approval-relevant, legal and safety-relevant requirements, fundamentally important functions, known and specified BM and BM from lessons learned in previous projects.

2 System DFMEA

Perform the DFMEA for the system. The objective is to define the meaning of the failure impacts from the customer's point of view. Look at the last level (component/subsystem) of the system/subsystem structure DFMEA.

3 Concept filter - defining the BFF

Identify the components that have a relevant impact from the customer's point of view in the case of a failure. The BFF are defined in this way. Additionally check according to GS 90017 whether rel. characteristics and rel. quality characteristics with an $S \geq 7$ shall be adopted as Special Characteristics.

4 Handshake between system FMEA and component FMEA (SW/HW)

Reconcile the BFF, PBM and the existing BMs with their severities S with the process owners of the subordinate FMEAs.

5 Component FMEA

Perform the component DFMEA. The objective is to identify the characteristics that have a significant impact on the BFF. Known and specified Special Characteristics from predecessor projects shall also be considered.

6 Design filter - defining the Special Characteristics

Decide whether potential Special Characteristics shall be further maintained as Special Characteristics or may be dropped. The Special Characteristics filtered out shall be marked and justified in the FMEA. BM-L that are relevant to type approval shall only be filtered out with the approval of the specialist department "Type Approval, Functions, Components, Motorcycle".

7 Handshake between component DFMEA and PFMEA

Fine tune the Special Characteristics and their severities S with the process owner of the PFMEA to ensure continuity of the Special Characteristics. The part drawing (poss. also single part drawing) is authoritative for the downstream process steps.

8 Production filter

Depending on the location of implementation (specified on the drawing), it shall be determined for the respective production or assembly step whether or not the Special Characteristic is relevant here. If so, suitable actions shall be taken (e.g. safe production sequences, reliable discovery).

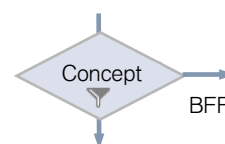
The Special Characteristic is kept on the drawing in all cases.

9 FMEA process and process control

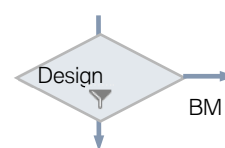
Perform the PFMEA.

For the risk assessment, the expected probability of occurrence in the process shall be evaluated here, from which the occurrence O of the PFMEA is defined.

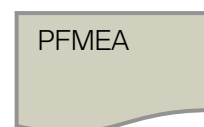
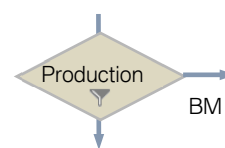
Transfer the BMs to be checked to the production control documentation, e.g. PLP or test schedule.



Handshake



Handshake



6.2 Identification of Special Characteristics without an FMEA

If this leads to Special Characteristics being identified, a PFMEA is always recommended in the process.

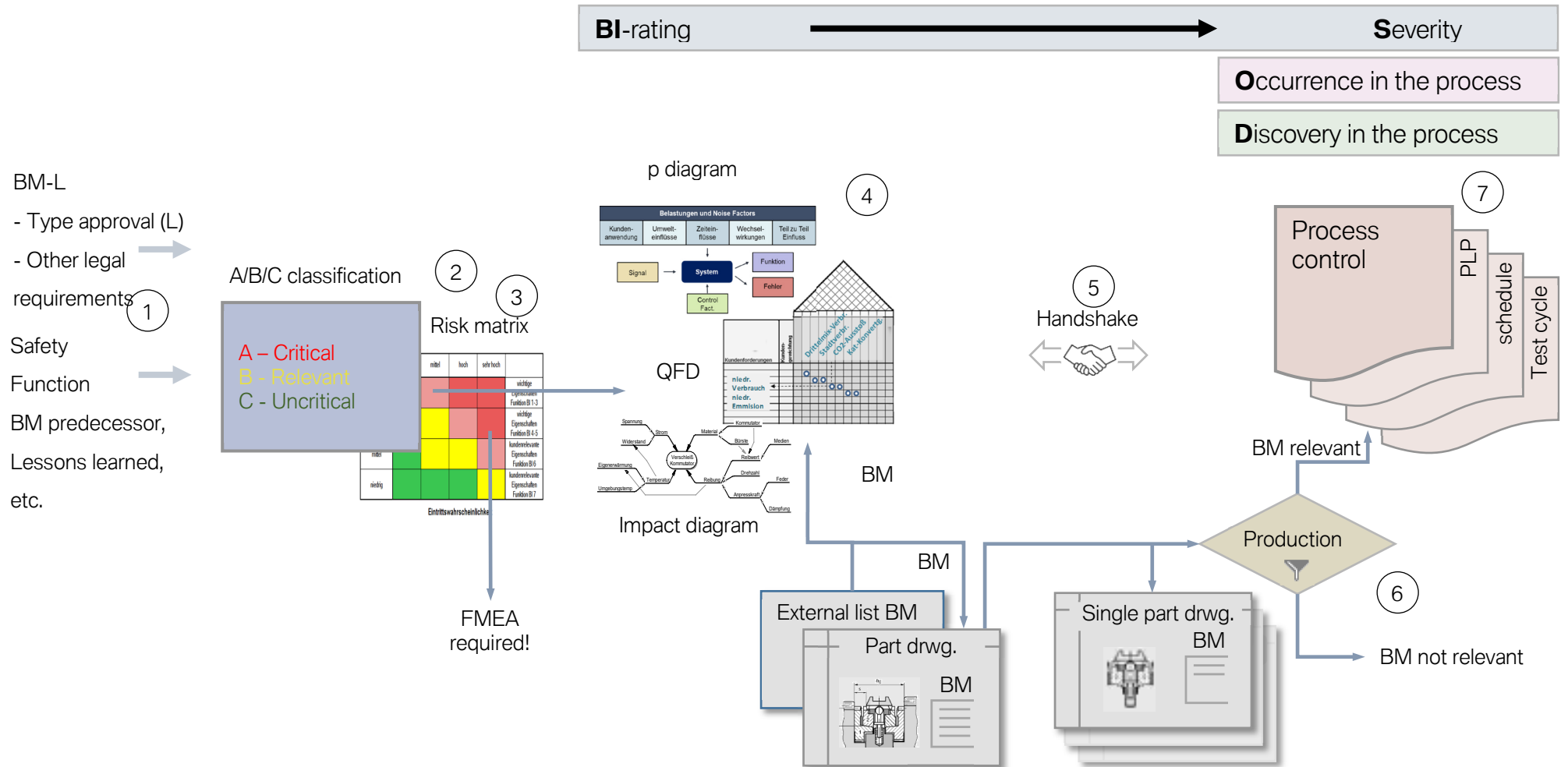


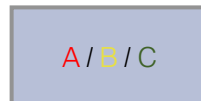
Figure 2: Alternative method for identifying BM without an FMEA in development

1 Requirements and specifications

Compile type approval-relevant components (see section 4.1), legal requirements, safety-relevant requirements, fundamentally important functions and known and specified Special Characteristics and Special Characteristics from lessons learned in previous projects.

2 A/B/C classification

Perform an A/B/C classification for the system/subsystem; see Annex E.8. For a classification A, the approach with a FMEA is mandatory. For a B, the approach with a FMEA is recommended. However, if this is not used, the fact shall be documented and justified.



3 Risk matrix (risk filter)

Using the risk matrix, identify the components that have a very high impact (BI 1 to BI 4) and a very high probability of occurrence and therefore have a very high risk.

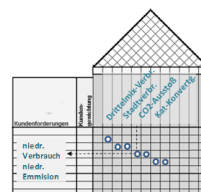
Check whether the malfunction of the component may be excluded as the failure cause by changing the concept (e.g. redundancy, safe fall-back level, fail-safe mechanisms), one of the following criteria shall be met:

- physical or logical impossibility (low probability of occurrence);
- robust characteristic (low probability of occurrence) according to sensitivity analysis (e.g. system analysis, virtual and experimental DoE, Monte Carlo simulation);
- proven design over many years with documented positive performance history without Special Characteristics under identical operating conditions (evidence from delta analysis).

	niedrig	mittel	hoch	sehr hoch	
Auswirkung	sehr hoch	schwerwiegend Fehler BI 1,2	schwerwiegend Fehler BI 1,2	schwerwiegend Fehler BI 1,2	schwerwiegend Fehler BI 1,2
	hoch	schwerwiegend Fehler BI 1,2	schwerwiegend Fehler BI 1,2	schwerwiegend Fehler BI 1,2	schwerwiegend Fehler BI 1,2
	mittel	schwerwiegend Fehler BI 3,4	schwerwiegend Fehler BI 3,4	schwerwiegend Fehler BI 3,4	schwerwiegend Fehler BI 3,4
	niedrig	schwerwiegend Fehler BI 5,6	schwerwiegend Fehler BI 5,6	schwerwiegend Fehler BI 5,6	schwerwiegend Fehler BI 5,6
	sehr niedrig	schwerwiegend Fehler BI 7	schwerwiegend Fehler BI 7	schwerwiegend Fehler BI 7	schwerwiegend Fehler BI 7

4 System analysis e.g. with p diagram, QFD and impact diagrams

Create a p-diagram (see example in Annex E.6) and/or a QFD. The QFD has the great advantage of being able to consistently derive BMs from the requirement up to the process. On the detail level, the impact diagram supplements the determination of the Special Characteristics (see example in Annex E.5).



5 Handshake between development and production process

Hand over the Special Characteristics determined using alternative methods to the production process and document this. The single part- drawing is authoritative for the downstream process steps.

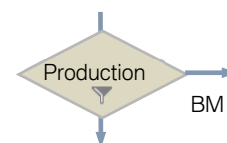
Handshake



6 Production filter

Depending on the location of implementation (specified on the drawing), it shall be determined for the respective production or assembly step whether or not the Special Characteristic is relevant here. If so, suitable actions shall be taken (e.g. safe production sequences 7, reliable discovery).

The Special Characteristic is kept on the drawing in all cases.



7 Process control

For the risk assessment, the expected probability of occurrence in the process shall be evaluated here, as per the occurrence O in the PFMEA. If BM-L and BM-S are transferred to the process and no PFMEA is carried out, this shall be justified by the responsible QM specialist department.

Transfer the BMs to be checked to the production control documentation, e.g. PLP and test schedule).



6.3 Explanations on the identification of Special Characteristics

6.3.1 Relationship between severity S or BI for categorization of Special Characteristics

BM-L has a severity of $S = 9$ or a $BI = 1$

BM-L has a severity of $S = 10$ or a $BI = 1$

BM-F typically has a severity of $S = 7$ to 8.

NOTE a BM-F may also have a lower S rating, for example, in the case of unwanted noise resulting in high warranty costs.

The breakdown in tabular form may be taken from Annexes E.1 or E.2

6.3.2 Applying the filters with and without a FMEA

The filters use a defined set of questions to support the identification of scopes that are actually relevant and shall be marked as Special Characteristics. The filters may also be applied during the identification of Special Characteristics with a FMEA. BM-Ls that relate to type approval may not be filtered out and are kept throughout.

Explanations and examples relating to filters are provided listed in:

- Annex B.1 Concept filter (system/subsystem)
- ;
- Annex B.2 Design filter;
- Annex B.3 Production filter.

If questions from the filter characteristics lead to a classification of no longer relevant, this shall be documented and justified in the FMEA and in RIMA.

7 Application of Special Characteristics in production

7.1 Defining process measures

If a production or assembly step in production has an influence on the BM, then measures shall be defined to avoid deviations and to discover deviations. The avoidance and detection measures shall be documented as tests in the production control documentation (e.g. PLP) after optimization within the framework of the PFMEA.

For all BM, the process capability according to GS 98000 shall be determined, verified and requalified during series production.

All BM shall be manufactured in one capable process. If this is not possible, 100 % validation shall be provided. Capability is considered capable when the following values are achieved for:

- BM-L/-F: $C_{pk} \geq 1,33$;
- BM-S: $C_{pk} \geq 1,47$.

For BM that may only be tested destructively, 100 % testing is not possible and a C_{pk} value is difficult to determine (see also GS 90017). In this case, process models shall be created to predict the characteristics and control them using process parameters. In addition, batch-wise destructive random sample inspection shall be performed.

7.2 Handshake with downstream processes

If validation of the BM is not possible within the process, then the BM shall be handed over in the handshake to the validating process (e.g. EoL, joint testing). The location of implementation shall be documented in the summarizing table of the BM for the component.

NOTE This handshake is not shown in Figure 1 and Figure 2.

8 Designating Special Characteristics and L-marking

Components, functions and overall vehicle characteristics for which Special Characteristics (BM-L, BM-S, BM-F) have been identified shall be identified with respect to these BM in documents (e.g. technical regulations, drawings, release documents, lists) and systems. This identification takes the form of L-marking or stating the characteristics category.

At BMW, characteristics without relevance to homologation are recorded in the specifications and drawing documents; they are not reported in quantity structures. Differentiation between other legal and type-approval-relevant characteristics is achieved by stating the application case, which enabling a clear assignment.

Change management in the scope of approving change management, as per the type-approval relevant scopes does not take place.

8.1 Denoting Special Characteristics in summarizing tables

In technical regulations and release documents in development and on (external) tables and drawings, all designated BM shall be indicated as BM-L, BM-S or BM-F in a summarizing table with unambiguously countable enumeration. The summarizing table is the central specification for the BM in all documents and states the category and application case.

The summarizing table of Special Characteristics contains at least:

- the application case;
- the BM category;
- the characteristic;
- the specific objective;
- the requirement;
- the location of implementation.

Further information may be added as required.

The technical implementation shall be documented by the specifying specialist department.

Changes shall be documented by the specifying specialist department and agreed upon with the releasing specialist department in advance.

Table 2: Designation of the Special Characteristics on a drawing or in the 3D model (example)

Application case	BM			Characteristic	Purpose	Requirement	Location of implementation ²⁾	Drwg. Pos.
	-L	-S	-F					
Type approval (L)	L1			Manufacturer's name	relevance to type approval/ product conformity	embossing depth ≥ 9 mm	releasing specialist department BMW / production supplier	S8
Type approval (L)	L2			Part number	relevance to type approval/product conformity	embossing depth ≥ 9 mm	releasing specialist department BMW / production supplier	S9
Type approval (L)	L3			Raised marking	relevance to type approval/product conformity	0 ≥ raised marking ≤ 1,0 mm	releasing specialist department BMW / production supplier	M14/M1 5
other legal requirements (G)	L4			Labelling of laser classification	legal requirement	Labelling of the laser classification shall be in the language of the country of use	releasing specialist department	
Safety		S1		Wall thickness of profiles relevant to crash cases	Ensuring safety requirements (crash)	4,5 ^{+0,3} _{-0,4} mm	production supplier	S8
Safety		S2		electric strength	Ensuring safety requirements	AC electric strength for TYPE 1 V > 50 V- 300 V	production supplier	A2
Safety		S3		Welding width	Ensuring safety requirements	Circumferential ≥ 3,7 mm	production supplier	C5
Type approval (L)	L5	S4		Wall thickness ¹⁾	Ensuring safety requirements	2,7 ± 0,3 mm	production supplier	D4
Safety		S5		Technical cleanliness	Ensuring safety requirements (contact gap)	maximum permissible contamination (size and number of particles as per table)	production supplier	
Function			F1	Parallelism delivery module	Ensuring level measurement	≤ 5,0 of reference level S to Z, process capability according to GS 98000	production supplier	E1
Function			F2	Leakage rate	Ensuring the leak-tightness of the part	≤ 1 x 10 ⁻³ mbar l/s at p = 150 mbar	production supplier	A1
Function			F3	Flatness	ensuring engraving capability	< 100 µm	production supplier	F3
	5	5	3					

- 1) A characteristic may belong to the characteristics category BM-L and/or BM-S or only to the characteristics category BM-F, see example "Wall thickness".
- 2) The detail in location of implementation is important to ensure that the process owning area is defined in case of a check of the Special Characteristic. The possible locations include Development, BMW Purchasing department, supplier, Production, Assembly.

The summarizing table of BMs shall be maintained (internally for BMW) in PRISMA for each part number. It contains the Special Characteristics for all release documents (e.g. 3D model, Office supplement, drawing) that are maintained with this part number.

NOTE Until the implementation of the summarizing table in PRISMA is available, an appropriate summarizing table of Special Characteristics is a constituent part of the release documents; it is maintained in the 3D model or in the drawing. If there is insufficient space on the drawing sheet, the summarizing table of Special Characteristics may be maintained in a separate document. An appropriate reference shall be stated in the drawing.

For applicable documents (e.g. quality specification), the Special Characteristics are stated in a separate summarizing table of Special Characteristics in the appropriate document.

8.2 Designating L-marking and type-approval relevant BM-L in quantity structures and change management systems

The L-marking and the BM-L for type approval (L) are shown in the relevant quantity structures and possibly the downstream systems; these are currently:

- for components, the generic part structure from PEP PDM (including PRISMA and the product master data);
- for electrical system/electronics functions, the AIDA FL;
- for data relevant to emissions and relevant to consumption (EVE) and complete vehicle characteristics (OtD) the SLA concluded with the responsible specialist department;
- possibly further quantity structures.

The L-marking and the BM-L relevant for type approval (L) are shown for components, electrical system/electronics functions and complete vehicle characteristics in the change management systems and their downstream systems; these are currently:

- for components and possible complete vehicle characteristics, M-gAMS;
- for electrical system/electronics functions, SWL-C.

8.3 Designation in drawings, graphics and 3D models

The BMs shall be marked in drawings, the graphics content of text documents and 3D models. The characteristics categories BM-L, BM-S and BM-F with unique, consecutive numbering for each characteristics category are specified before the characteristic. A semicolon is inserted between the numbering and the characteristic. Zeppelin refers to the elliptical frame around the feature indication. It serves as a visual highlight. The zeppelin frame is black, the background colorless; see Figure 3.

In drawings, the Special Characteristics shall be marked with a "zeppelin".

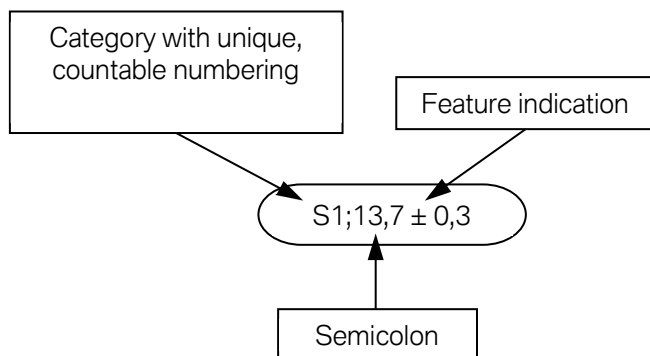


Figure 3: Marking of Special Characteristics in drawings with a "zeppelin" (example)

Consistency from the FMEA to the drawing shall be ensured. However, for existing projects the previous markings cc/L, cc/S and sc/F may still be used.

If direct marking of the characteristic with a zeppelin is not possible (e.g. in the case of geometric tolerance, surface roughness, texts), the characteristics category shall be marked with a zeppelin containing unique, countable numbering in the direct vicinity of the Special Characteristic; see Figure 4. The relationship shall be indicated by an arrow to the Special Characteristic. If the relation is unambiguous, the arrow may be omitted.

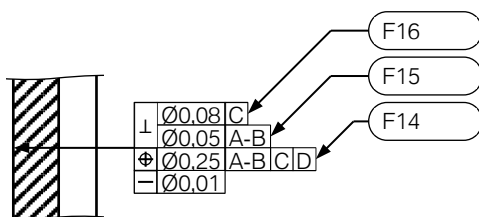


Figure 4: Marking of positional tolerances as a Special Characteristic

In the graphic content of text documents and in 3D models, the Special Characteristics may additionally be marked with a "zeppelin". The marking shall be made in a uniform manner within a 3D model.

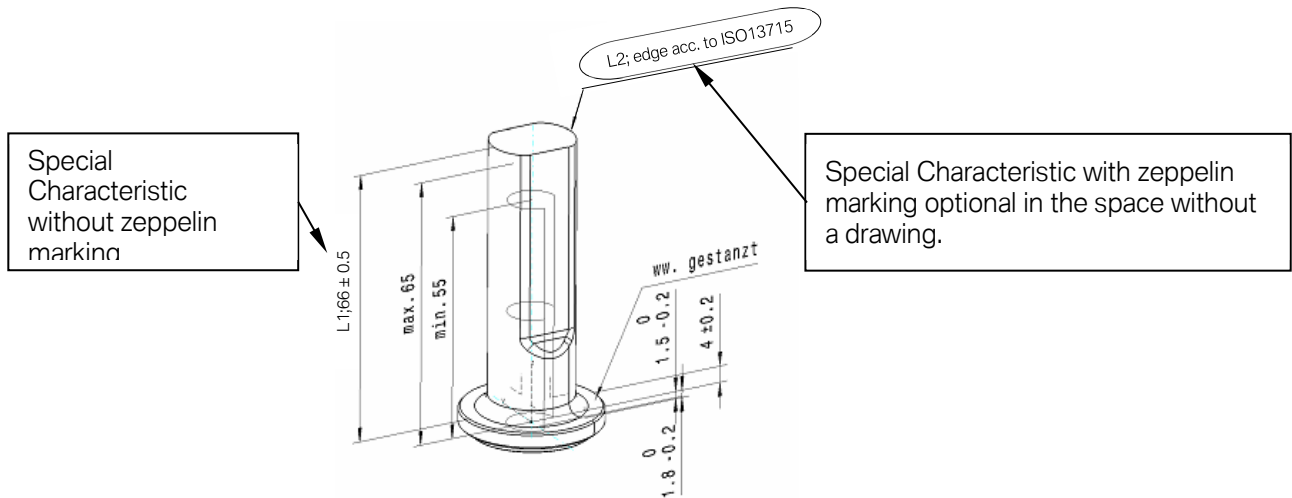


Figure 5: Example entry for Special Characteristic in 3D models

The BMs shall be stated the text header of the drawing.

						Methode 1 ISO 128				Bayerische Motoren Werke Aktiengesellschaft					
Oberflaechen: Form-und Lagetoleranzen:						VDA 2005 DIN EN ISO 1101				Alle Rechte vorbehalten/ALL RIGHTS RESERVED					
Rohteil Nr./Zwischenzustand: -				Lieferant: -						Erstellt mit/PRODUCED ON CATIA 5.14SP5P					
	Datum	Name		Datum	Name	Kennz. nach GS 91011 L5 S5 F3	Massstab SCALE DO NOT SCALE	Gewicht mit Toleranz/WEIGHT WITH TOLERANCE nach GS 90006			Blatt - von - Blatt				
Gez.	-	-	Fertigg. gepr.	-	-										
Gepr.	-	-	Werkst. gepr.	-	-										
Funkt. gepr.	-	-	Gesehen	-	-										
												Ersatz fuer -		Ersetzt durch -	

Figure 6: Marking in the text header of the drawing (example)

NOTE As long as the summarizing table of BM is not stored in drawing management (PRISMA), the number of BM per category shall be shown in the text header.

8.4 L-marking on the cover sheet of documents

The characteristics category L used shall be stated on the cover sheet of the technical documentation.

BMW Group Standard	Freigegeben	GS 93041-1 2018-04		
		Ersatz für GS 93041-1:2016-03		
<table border="1"><tr><td>L</td><td>Kennzeichnung nach GS 91011</td></tr></table>		L	Kennzeichnung nach GS 91011	
L	Kennzeichnung nach GS 91011			
Laminateigenschaften CFK-Karosserie-Strukturbauteile Allgemeines				
Bearbeiter, Normung Martin Ruppel				
Ausdrucke unterliegen nicht dem Änderungsdienst. Der Anwender dieses Dokuments ist verpflichtet sich über den gültigen Stand zu informieren.				
Fortsetzung Seite 2 bis 47				
BMW GROUP © BMW Group - Alle Rechte vorbehalten	  			

Figure 7: L-marking on the cover sheet

9 Documentation and archiving

The use and retention periods of documents and records relating to the BM are as follows, based on the recommendations from VDA Volume Special Characteristics:

- 30 years for BM-L and BM-S;
- 10 years for BM-F.

A (Annex informative) Technical categories of Special Characteristics

From the technical point of view, the BMs are broken down as shown in Table A.1 with their possible tests.

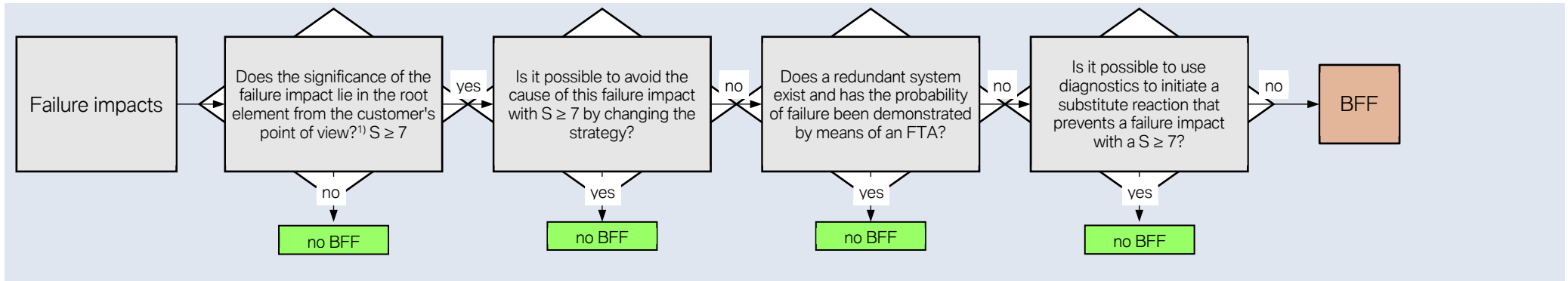
Table A.1: Overview of technical categories of Special Characteristics with examples

Type	Type	BM (Examples)	Test (Examples)	Test type (Examples)
attributive	Existence	– label – vehicle identification number	Camera test	100 %
	Function (yes/no)	– light – current – snapped-in	Function test	100 %
Ordinal	countable	– number of teeth – number of revs	Digital count	100 %
Categorical	variant (A, B, C, etc.)	– component variant – country-specific variant – manufacturer – material type	(camera test) bar code, OCR.	100 %
Quantitative	characteristic	– geometry – surface	Process capability	$C_{pk} / 100 \%$
		– material composition	material test	first/last part release
	function, continuous	– current/torque	characteristic curve	$C_{pk} / 100 \%$
	Quality characteristic	– strength – hardness	tensile strength test	Random sample
	Service life	– cycles – operating hours	Stress test	Random sample
	technical cleanliness	– number/size of particles	technical cleanliness test	UQL ¹⁾
1) as a substitute value for evaluating the technical cleanliness process capability; overshoots in percent				

B (Annex normative) Filters

B.1 Concept filter (system/subsystem)

The objective of using a concept filter is to identify the components that have a relevant impact from the customer's point of view in the event of a failure and whose BFF shall be transferred to the lower-level FMEA or to the documentation of alternative methods in the handshake.



1) The topmost element in the FMEA structure tree.

Figure B.1: Summary of strategy filters

If the questions shown in Figure B.1 do not lead to a BFF, then this shall be documented in the FMEA by an appropriate attribute.

EXAMPLE Filter 2 light sensor: A sensor switches the vehicle's light. If this fails, the severity for the light actuation error path is $S = 10$. The strategy is now changed so that the light is switched if the sensor does not supply a signal (safe fallback level). The severity for the light continually switched on is only $S = 6$.

B.2 Design filter

The objective of using a concept filter is to identify the characteristics that have a significant impact on the BFF and pass them on to the downstream processes as BMs.

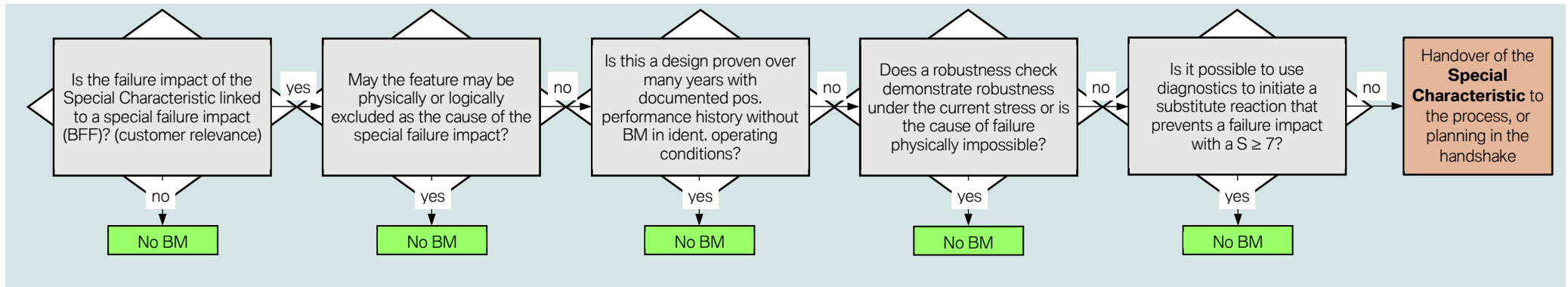


Figure B.2: Summary of design filters

If the filter does not lead to a BM, then this shall be traceably documented with the corresponding justification.

B.3 Production filter

In contrast to the previous filters, the production filter does not reduce the number of Special Characteristics. All BMs determined up to this stage are adopted into the part drawing in all cases.

Depending on the location of implementation (specified in the summarizing table for the Special Characteristic in the drawing), it shall be clarified for the respective production or assembly step whether the Special Characteristic may be influenced here and is thus at all relevant. If not, then it shall not be taken into account in the respective production control documentation (e.g. PLP) or in the test schedule.

If further assembly or further process steps are not possible, the defect is directly discovered by this. In this case, no additional tests are needed and the Special Characteristic is not relevant here.

This shall be documented and justified in the PFMEA.

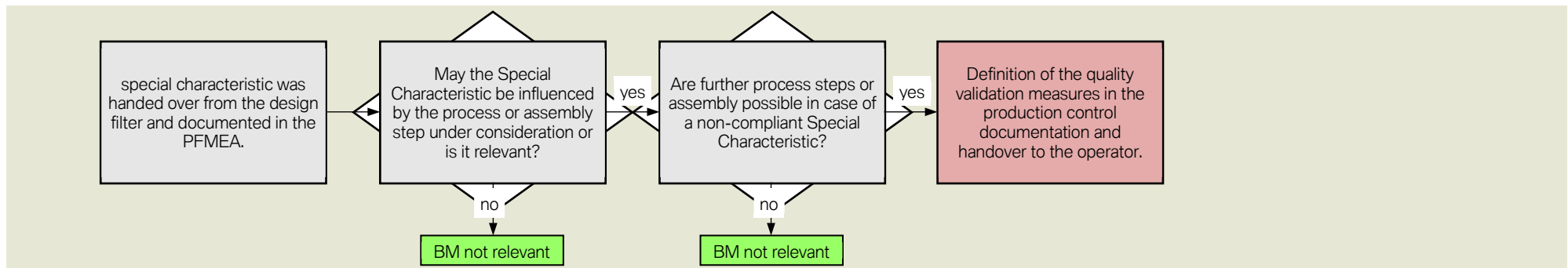


Figure B.3: Summary of production filters

Supplementary consideration for production

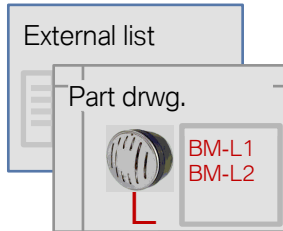
The failure impacts of the Special Characteristics exclusively relate to the customer view. However, it is possible in production or assembly that, e.g.:

- there is a risk of injury or that occupational safety is generally endangered;
- hazardous substances are handled;
- there is a risk of fire.

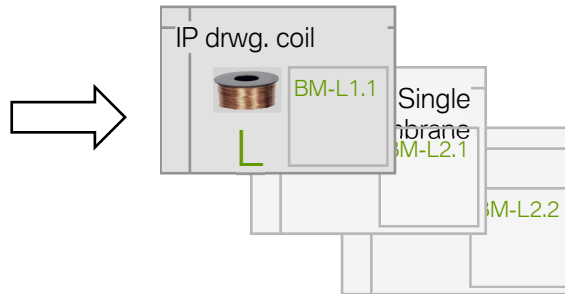
This and economic concerns, such as high rework costs, shall also be taken into account in the production control documentation (e.g. PLP) and in the test schedule.

C (Annex informative) Example decomposition

assembly drawing



single part drawing
production drawing
sub-component



Component drawing (assembly drawing)

Basis between BMW and supplier.

L-marking in drawing header

List of BM-L for the requirements,

EXAMPLE horn

- BM-L1 = volume
- BM-L2 = frequency

Only these BM-L are included in PEP-PDM, in PRISMA in the quantity structures and in change management.

These BM-L shall not be filtered out (the location of implementation may be, however, by the process-relevant characteristics, describe don the right).

Single part drawings/subcomponents

Basis for manufacturing at the supplier or in-house production.

L-marking in drawing header

List of BM-L process-relevant characteristics,

EXAMPLE horn

- BM-L1.1 = coil size (main influencer of volume)
- BM-L2.1 = membrane thickness (main influencer of frequency)

These BM-L are integrated into the production control documentation (e.g. PLP) and they are only included in the summarizing table on the drawing.

D (Annex informative) Overall view concept/design/production filters

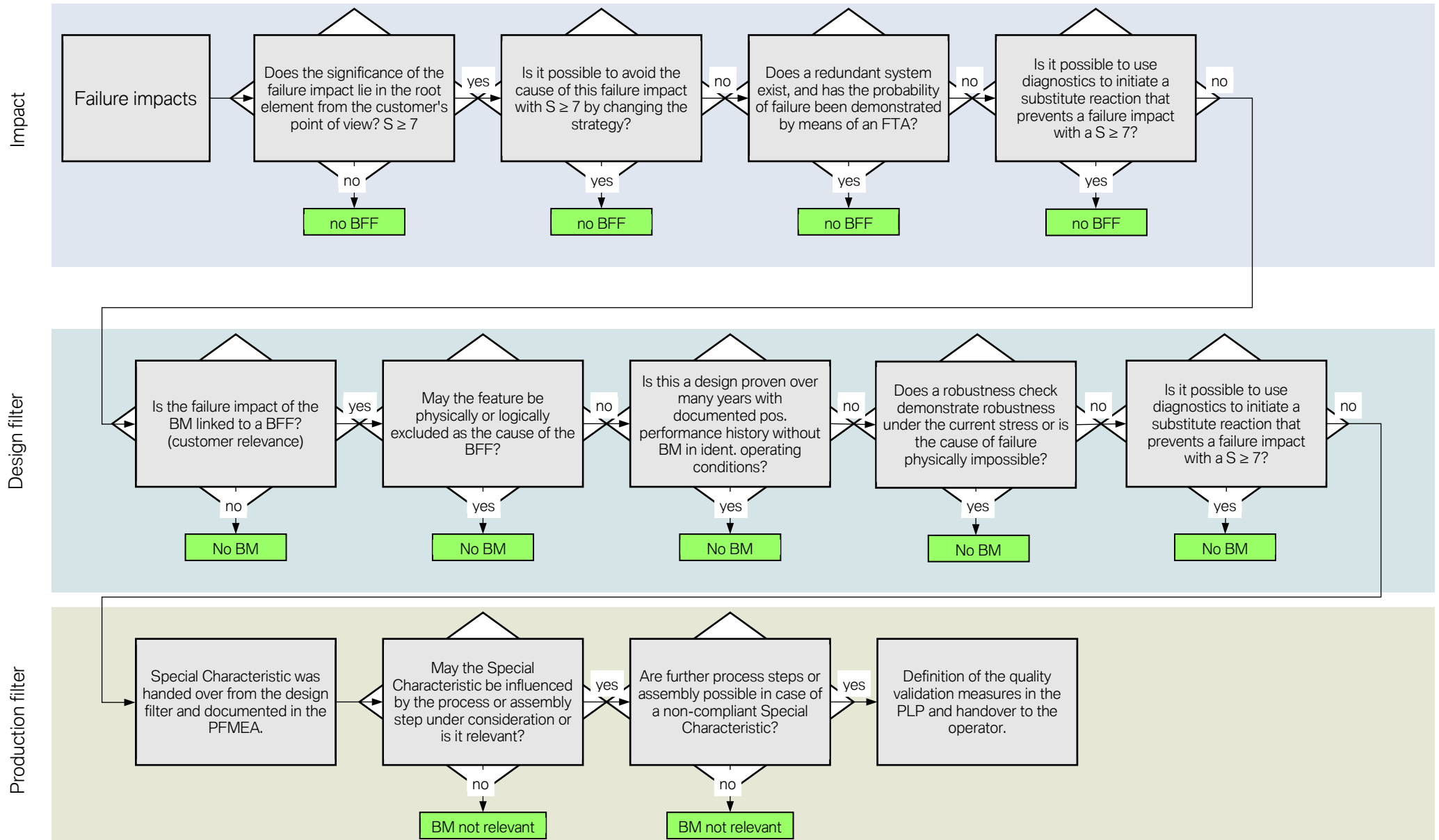


Figure D. 1: Overview of filters

E (Annex normative) Severity S and rating index BI

E.1 Relationship of severity S to Special Characteristics from the FMEA

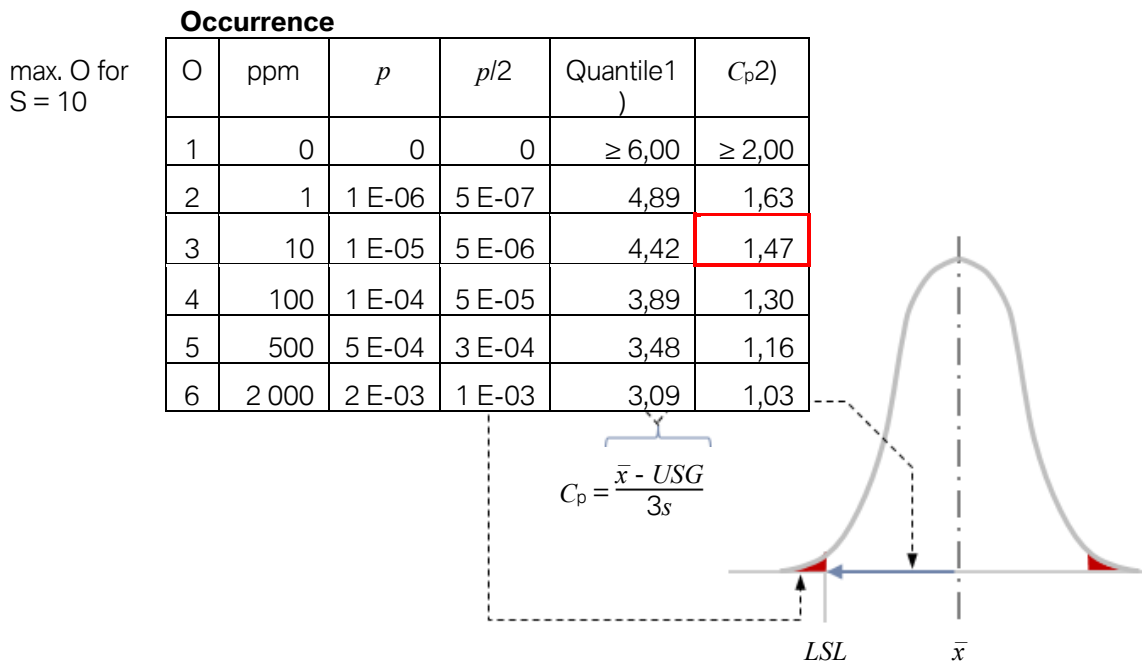
S	Severity	BM ¹⁾
10	Impact on the safe operation of the vehicle and/or other vehicles, the health of the driver or the front passenger or other road users.	S
9	Noncompliance with legal requirements and requirements from authorities	L
8	Loss of a major function required for normal vehicle operation over the intended service life	F
7	Restriction of a major function required for normal vehicle operation over the intended service life	
6	Loss of a comfort function	
...	...	

E.2 Relationship of rating index BI to Special Characteristic

BI	Rating index	BM ¹⁾
1	Safety defect posing a safety risk for the driver/environment	S
	approval-relevant topic	L
2	Product causes extreme annoyance e.g. due to failure; customer threatens legal action or publication	F
3	Product causes uncertainty, annoyance; customer immediately visits workshop	
4	Disappointment, defect leads to unscheduled shop visit	
5	Causes dissatisfaction on part of customer, unreasonable borderline case, customer has defect remedied during next shop visit	
...	...	

- 1) A Special Characteristic is not automatically assigned by the S and BI values displayed. A Special Characteristic is only a Special Characteristic if it has not been ruled out by one of the filters shown. It is possible for a S = 10 not to be a Special Characteristic as it was ruled out by the production filter. BI values may not be compared one against one with the severity S (guide values only). BM-F may also be assigned for a S < 7 or BI > 5. For example, this is the case with noise problems that may cause increased warranty costs. In the running gear area in particular, the BI-ratings may deviate from the definitions shown.

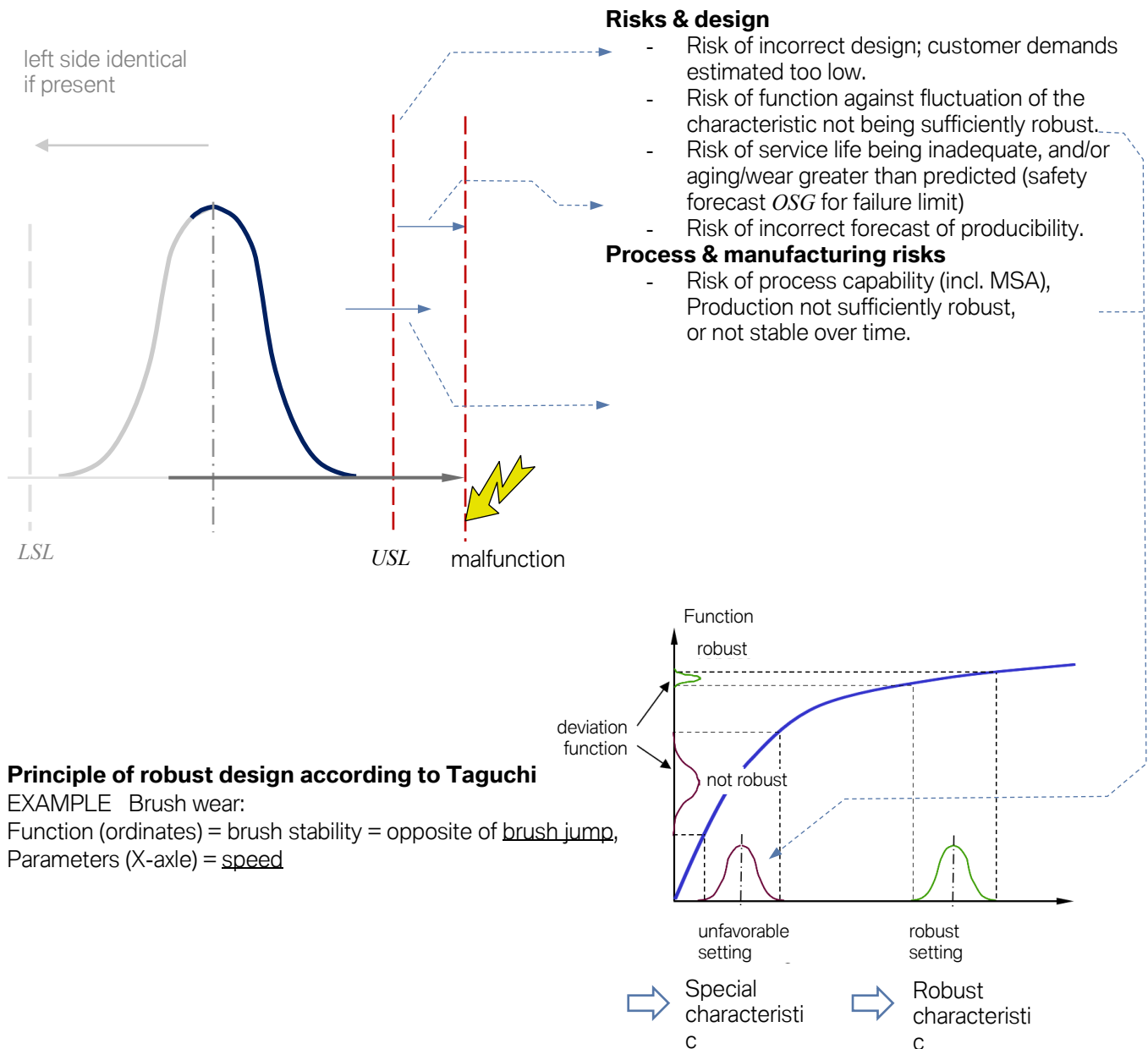
E.3 Probability of occurrence O for PFMEA and C_p



- 1) = -NORM.S.INV($p/2$)
- 2) = Quantile/3

NOTE In the figure, only the reference to C_p is shown for simplification purposes, to show the two-sided overshoot proportions symmetrically. The actual arrangement refers to C_{pk} ; see GS 98000.

E.4 Risks for development and the process for steady characteristics

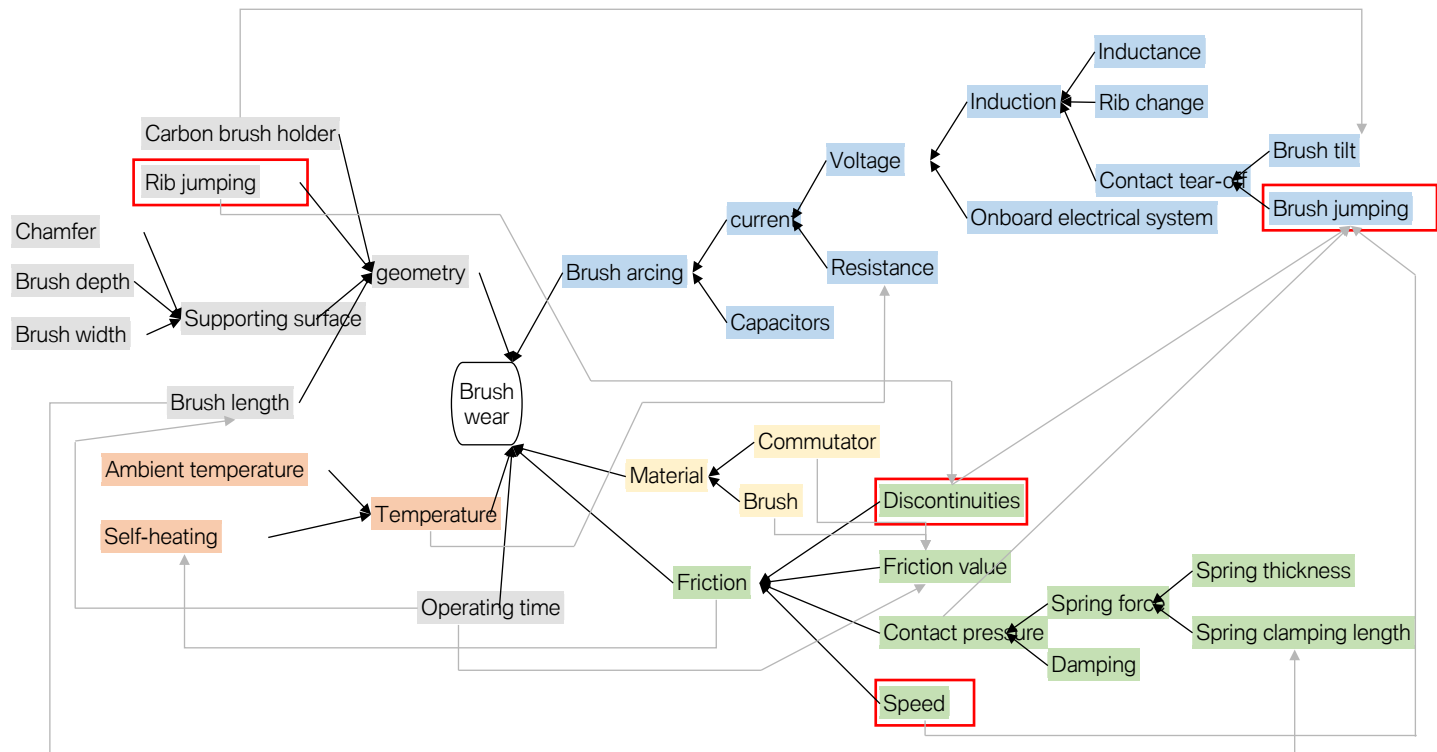


The principle of robust design according to Taguchi states that for a freely selectable range of a parameter that is difficult to control (here speed), the setting at which the most favorable function is produced shall be sought. Due to the known functional relationship (blue curve), this is the flatter upper range, as the deviation does not have such a strong effect here. This is, at the same time, the higher value of brush stability.

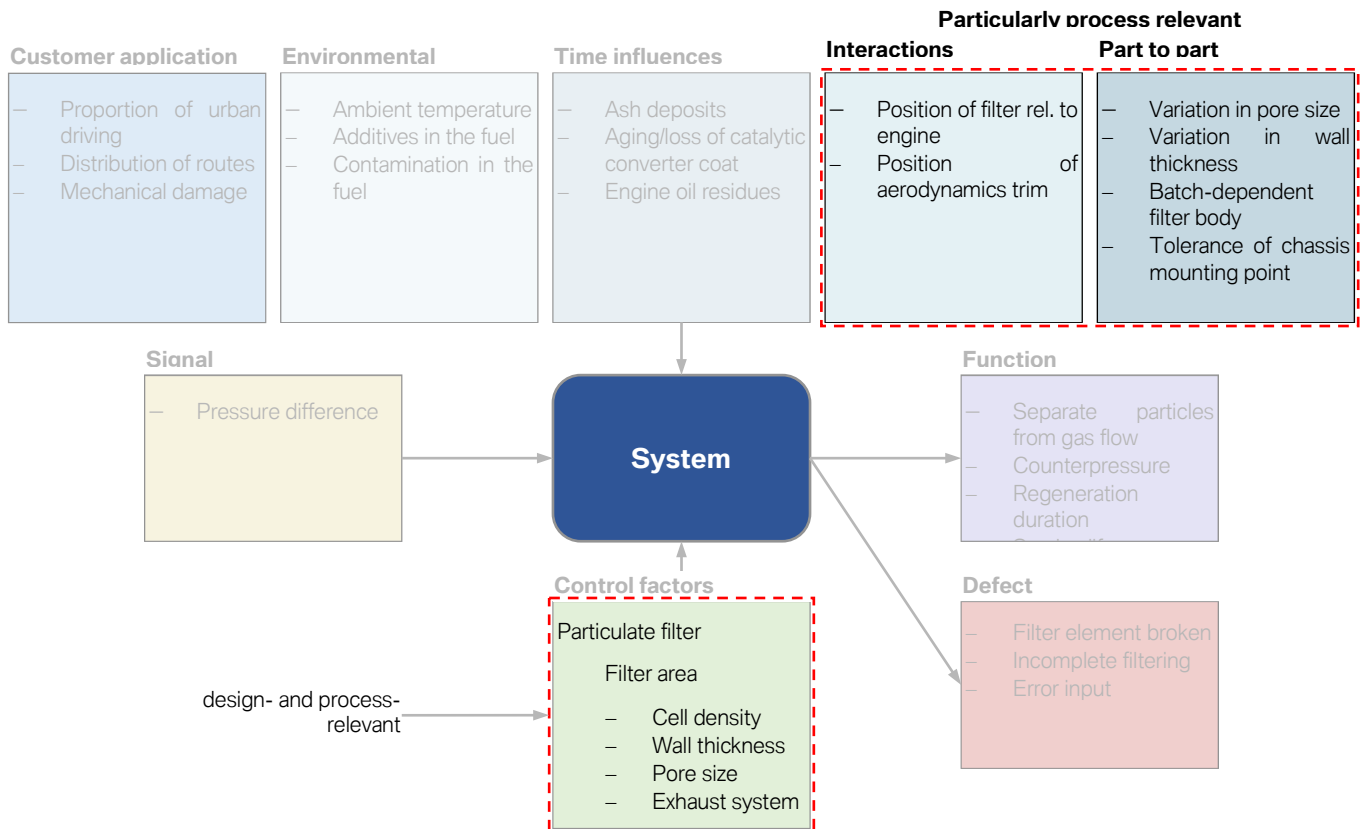
A problem occurs if the actual objective is a smaller value. In this case, the greater deviation shall be accepted, a robust setting is not possible in this case.

E.5 Example of cause and effect diagram for determining potential Special Characteristics without a FMEA

The following example relates to the wear on the brushes of an electric motor. Based on the found correlations about the failure cause brush arcing ... brush jumping ... discontinuity, the BM-F rib jump, in particular, was identified in addition to other characteristics. Here, the manufacturer shall demonstrate a maximum permissible upper tolerance via a process capability of $C_{pk} = 1,33$.



E.6 Example of P diagram and QFD for determining potential Special Characteristics/Special Characteristics without an FMEA

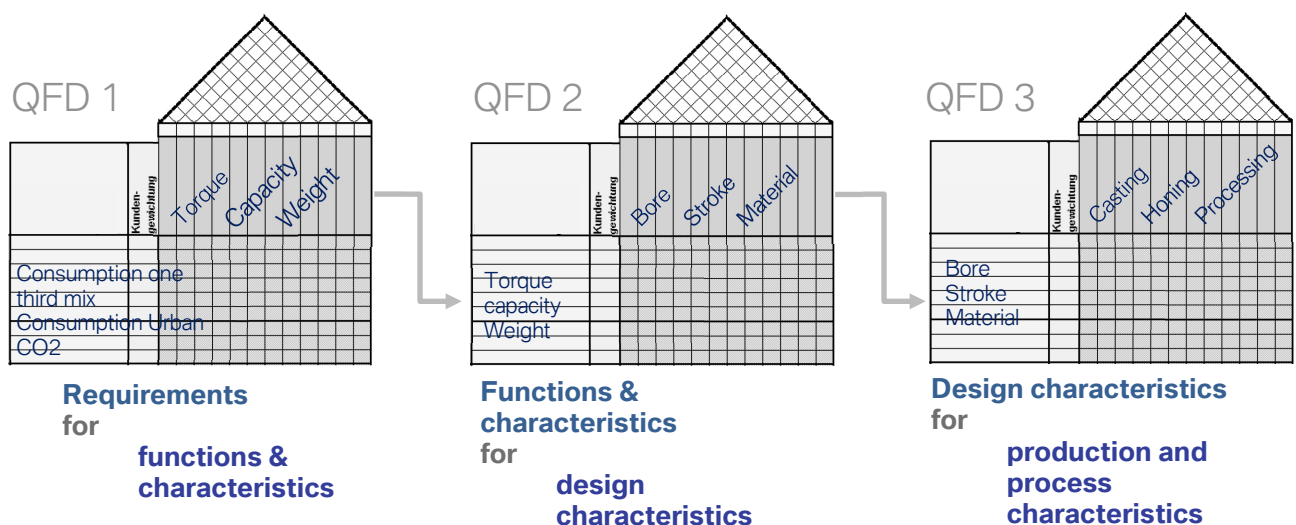


E.7 Basic principle of a QFD for determining potential Special Characteristics/Special Characteristics without an FMEA

QFD consistently represents the relationships between requirements, functions and characteristics (properties) and their production.

The following illustration shows a simplified version of this using an engine by way of an example:

Requirements → BFF → Characteristics BM → Process parameter



NOTE There are different approaches and breakdowns between the companies. Cause-effect diagrams, p diagram and other quality tools may ideally complement each other.

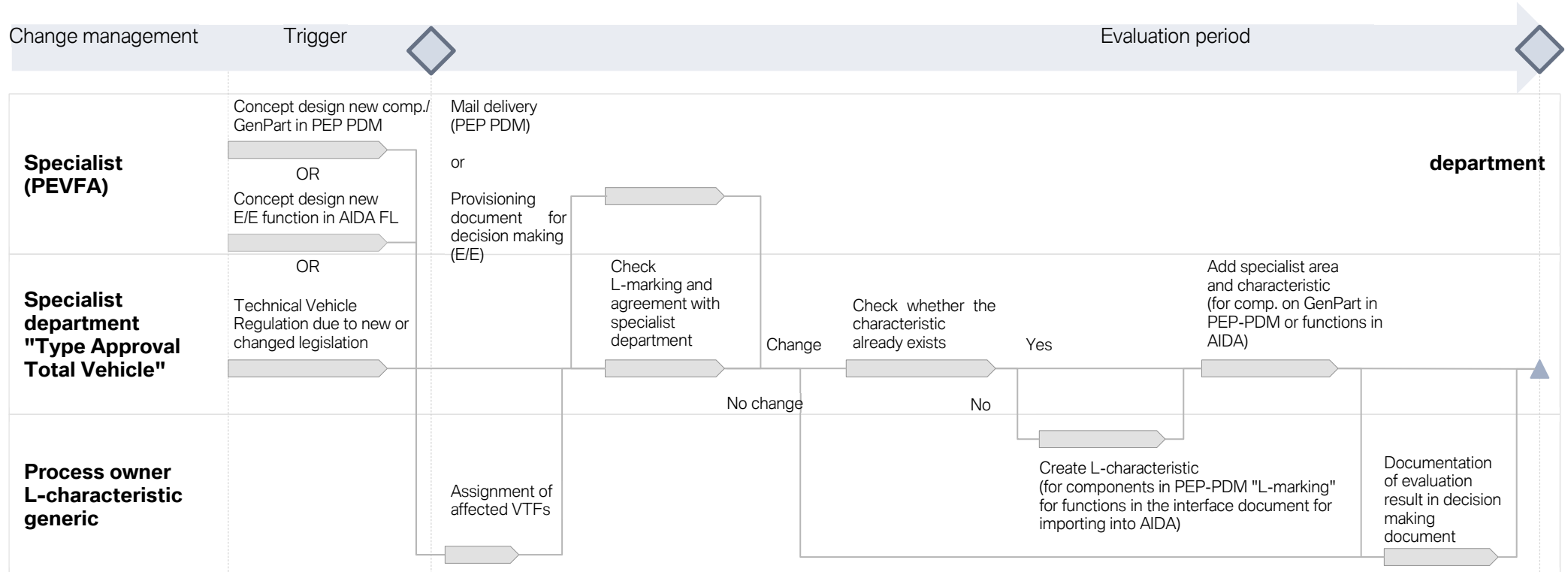
E.8 A/B/C risk classification

The A/B/C classification is a part risk assessment tool. A/B/C stands for a high, moderate or low risk class. The following table provides a schematic overview:

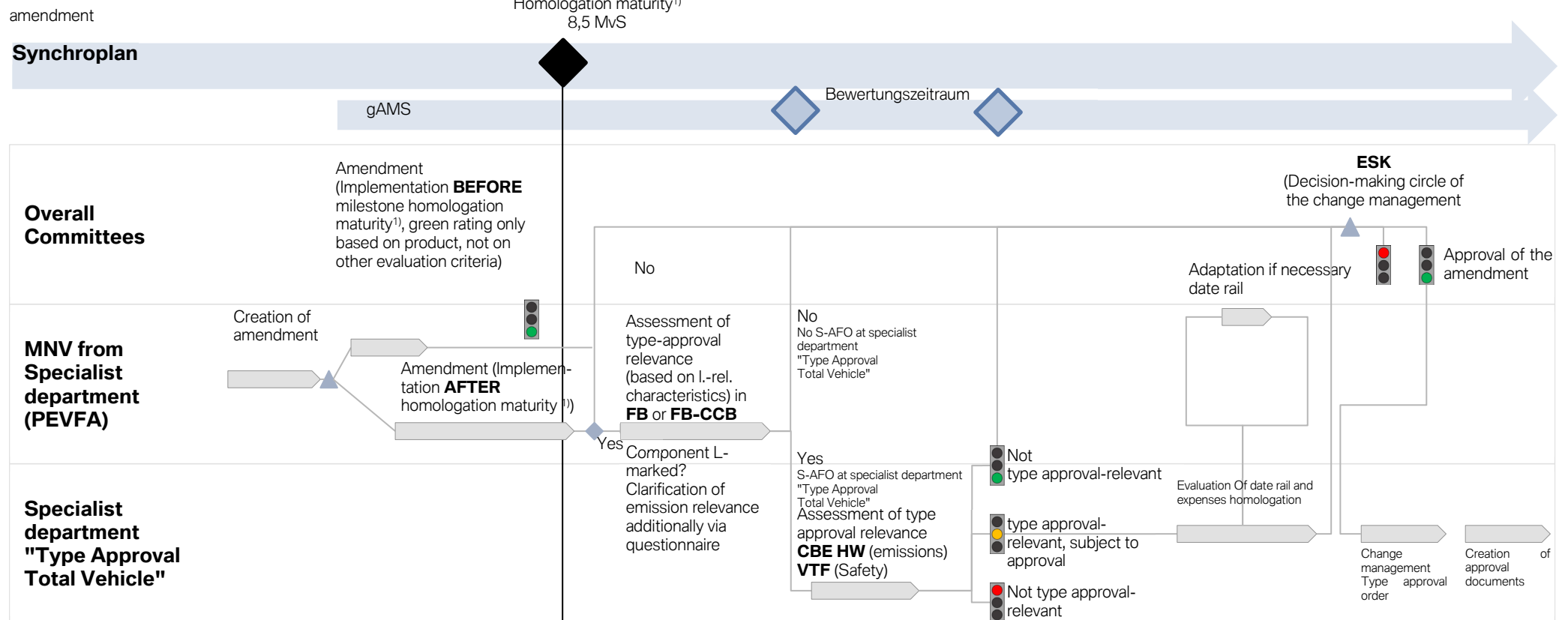
Classification	Severity	Consequences
O	Critical and thus highly relevant scopes for risk assessment EXAMPLES new concept, high complexity, high financial impact, safety- or approval-relevant	The application of the risk matrix and in case of a high risk assessment a subsequent FMEA is mandatory. This may only be omitted in consultation with line management (e.g. Quality Steering Committee, Prevention)
S	Relevant scopes for risk assessment Examples: concept only changed to a minor extent or carryover, minor irregularities of predecessor model, manageable complexity	
C	Non-critical scopes and thus no intensive risk assessment EXAMPLES known/proven concept, no prominent irregularities in the case of the predecessor model, however, requirements management not yet completed	In case of a carryover from a predecessor project, a delta assessment shall be performed at a minimum. There is no need for a FMEA.

F (Annex normative)

F.1 Adaptation/new assignment of L-marking and type approval-relevant BM-L



F.2 Change management for components after homologation maturity project milestone

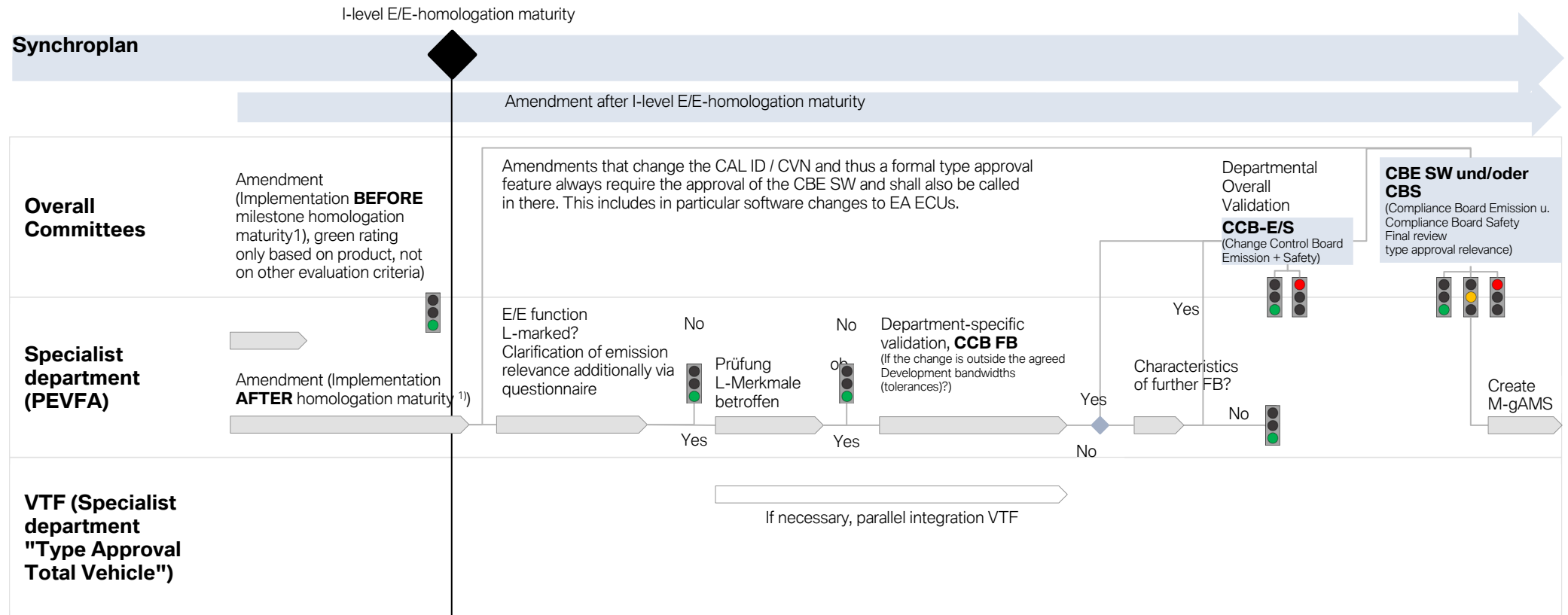


NOTE In principle, in the event of planned changes, specialist agencies should check for the relevance of the authorization of EG-8 as early as possible in order to enable rapid gAMS processing. Ideally, the examination will take place before an amendment is tabled.

- Implementation ordered**
- Change is subject to approval
- Implementation rejected

1) First launch; milestone deadlines may vary specifically for each project, see respective Flexlto synchronization plans for launch

F.3 Change management for electrical system/electronics functions after the homologation maturity project milestone



Implementation ordered

Initiation of standard type approval process (incl. finding deadlines and gAMS)

Implementation rejected